QPCODE **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) EXAMINATION

THIRD SEMESTER

COURSE CODE - MG3DSEMBG200

COURSE TITLE: BIOINSTRUMENTATION AND TECHNIQUES

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **1** mark.

1. Which part of the microscope is modified to achieve dark-field illumination? [K]/ [CO1]
2. What parameter determines the sedimentation rate in differential centrifugation?[U]/[CO2]
3. Expand PAGE. [K]/[CO3]
4. Name the law on which colourimetry is based. [K]/[CO4]
5. What type of primers are used in PCR? [U]/[CO5]

[1x5 = 5]

Part B

Short Answer Questions

Answer any **10** questions.

Each question carries **2** marks.

6. Give any two applications of fluorescence microscopy in microbiology. [A]/[CO1]
7. Define differential centrifugation. [K]/[CO2]
8. What is the function of agarose in AGE? [U]/[CO3]
9. Define spectrophotometry. [K]/[CO4]
10. Give two advantages of nested PCR. [A]/[CO5]
11. Why are fluorescent dyes important for confocal microscopy? [A]/[CO1]

12. What is zonal centrifugation? How is it different from isopycnic centrifugation? [An]/[CO2]
13. What is the function of the blocking step in Western blotting? [U]/[CO3]
14. Write the mathematical expression of Beer-Lambert's Law. [K]/[CO4]
15. How does RFLP differentiate between bacterial strains? [U]/[CO5]
16. Why is phase contrast microscopy useful for observing live cells? [U]/[CO1]
17. What is the first step in two-dimensional electrophoresis? [K]/[CO3]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **6** marks.

18. Explain the principle of resolution in microscopy. How does numerical aperture (NA) influence it? [A]/[CO1]
19. Explain the applications of density gradient centrifugation in molecular biology and virology. [U]/[CO2]
20. Define SDS-PAGE and explain its working principle. [U]/[CO3]
21. Explain how turbidometry can be used to measure microbial growth in broth culture [U/[CO4]
22. Briefly describe the basic steps of RAPD analysis. [A/[CO5]
23. Describe the basic working principle of SEM. [A]/[CO1]
24. Why is DNA fingerprinting useful in paternity testing? [A]/[CO5]

[6x5 = 30]

Part D

Long Essay Questions

Answer any **1** question.

The question carries **15** marks.

25. Explain the working principle of TEM with a simple diagram. [An]/[CO1]
26. A researcher wants to compare soil bacterial communities. Justify why RAPD would be appropriate. [E]/[CO5]

[15x1 = 15]